



DEALER BEST PRACTICE SERIES

Recommended Shipping
and Storage Stands for
Major Components

Application Maintenance Site Management Component Rebuild Safety MARC Management

Recommended Shipping and Storage Stands for
Major Components 1

1.0 Introduction 2

2.0 Best Practice Description 2

3.0 Implementation Steps..... 4

4.0 Benefits 5

5.0 Resources Required..... 5

6.0 Supporting Attachments / References 5

7.0 Related Best Practices 5

8.0 Acknowledgements 6

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1.0 Introduction

From the time a major component is rebuilt and leaves the CRC until it is installed in a machine, the component is handled several times and exposed to risk of damage each time it is handled. Typically, the component moves from the CRC shipping area on to a flatbed truck for transport to the customer, from the truck to a receiving and storage location, and from that location to a customer shop for installation.

A well designed, sturdy shipping and storage stand can protect the component from shipping damage and make the handling of the component from one place to another much easier and more efficient to perform.

Some dealers CRCs require that all components rebuilt in the shop be transported / stored on a dealer supplied shipping stand. These dealers often sell, rent, or lease the component shipping stands to their customers.

2.0 Best Practice Description

The methods and practices used to transport and store a major component vary widely. It is not uncommon to see a major component on a flat bed truck supported by a pallet or wooden blocks, and tied down with a few ropes and chains. Makeshift shipping and storage practices expose the component to potential shipping damage.

Caterpillar ships Remanufactured major components on custom shipping stands, providing a secure mount for the component and built-in channels to facilitate forklift handling. A shipping stand core charge is assessed with the sale of the Reman component and credited when the core is returned. The same stand that was used to transport the Reman component is used to return the core to Caterpillar. Some dealers have developed a similar practice between their CRC and customers. Many dealers use their own weld and fabrication shops to design and manufacture component shipping and storage stands.

Design features of the stands often include base channels to allow transport by forklift and a secure method to mount the component to the stand. Components such as engines are usually bolted to the stand using the same mounting locations used to install the engine in a machine. Round components, such as final drives and transmissions, are usually chained to a stand and supported by cradles sized to securely hold the component.

For engines, there are “universal” designs to fit several engine sizes by using different mounting adapters.

While shipping and storage stands are important to the transport and protection of major components, the stands cost time and money to manufacture. Some dealers have difficulty in getting stands returned in a timely manner or not getting stands returned at all. The solution is to sell, rent, or lease shipping stands. In this way, the cost of the stands can be recovered and the shop can afford to invest in a sufficient quantity of stands to assure that all components are properly shipped and stored on stands.

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Recommended Shipping and Storage Stands for Major Components	DATE 4/12/2017	CHG NO 02	NUMBER 0107-4.4-1053
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For cases where the customer requires a permanent set of component stands for an inventory of components, stands can either be purchased or leased from the dealer.

For situations where the stand is only required for shipping and temporary storage, there can be a refundable deposit on the shipping stand when the component rebuild is invoiced. If the component stand is returned within a specified period, the customer receives a full refund of the deposit. If the return of the stand goes beyond a set time period, a rental fee is applied.

Examples of shipping and storage stands:



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Recommended Shipping and Storage Stands for Major Components	DATE 4/12/2017	CHG NO 02	NUMBER 0107-4.4-1053



3.0 Implementation Steps

- Purchase finished component stands or design and fabricate stands to meet market needs. Designs are often similar to shipping stands used by Caterpillar Remanufactured Products or various assembly stands contained in Service Tool Guides.
 - Determine the numbers of stands required by looking at monthly CRC component rebuild volumes and average cycle time between shipping a component and receiving the stand back.
 - Designate serial or tracking numbers for each stand.
 - Warning decals to designate weight, lifting, and tie down instructions.
 - Keep engineering drawings for any fabricated stands so they can be replicated if needed.
- Create a tracking and invoicing system.

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Recommended Shipping and Storage Stands for Major Components

DATE
4/12/2017

CHG
NO
02

NUMBER
0107-4.4-1053

- Implement a policy requiring all components leaving the CRC to be mounted on shipping and storage stands.
- Communicate policy to customers and explain benefits.
- Create pricing and marketing program for purchase, rental, or leasing options of stands.

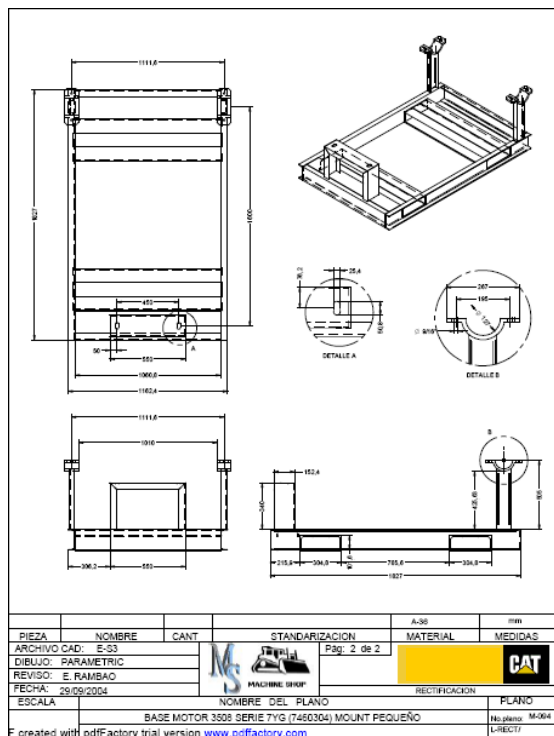
4.0 Benefits

- Stands prevent damage to components during shipping and handling.
- Improved material handling - safety and efficiency.
- Creates a standard, consistent process for the transport and handling of components.
- Conveys an image of quality from the CRC and dealer.
- Enables recovery of the cost of shipping and storage stands.

5.0 Resources Required

- Weld/fabrication shop
- Capital for materials and labor
- Tracking and invoicing system

6.0 Supporting Attachments / References



8.0 Acknowledgements

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Photos courtesy of Finning Chile, Gecolsa, and Venequip.

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